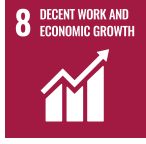




OPN (MAIN PLANT) Department  
SIMHADRI

**Note No :1**



**Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Date:**

24/11/2024(19:53:37)

**Subject:**LMI BARRING OF BARRING OF TURBINE GENEARTOR AND  
PREVENTION OF STALLING

LMI/OD/OPS/SYST/001 Issue No.: 1, Rev: 08, Date: Nov 2024

"BARRING OF TURBINE GENERATORS AND PREVENTION OF STALLING" has been revised as per revised OD Rev.No:5 Sep 2024

**Rev: 08 Dec 2024:** i)4.1 f added

ii)4.2.1.1.5 modified(seal steam trap checking frequency added)

iii)4.3.1.3.2 Recording of JOP back pressure changed to every 30 days from 15 days

iv)4.2.4.1 e Ensure MS/CRH/HRH line is in depressurized condition added

v)4.2.2.1 Incorporation of alarm of 1st st pr > 2ksc with turbine trip included

vi)4.2.6.3 added

vii)Proper instruction of stoppage of oil pumps in case of requirement is mentioned

viii) Aneexure-III check list added to counter reasons of barring and take corrective action in case barring stalling takes place

ix)4.2.2.10 Monitoring of brg temp during coasting down of turbine included and

monitoring of LO pr/temp,LO pr after barring gear valve,Condensor vacuum included

x)4.2.4.2 j added

No other changes made in this revision.

Put up for kind Approval

**Submitted by:**

किल्लि तिरुमल राजशेखर - उप महाप्रबंधक

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**On:**24/11/2024(19:53:37) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950)

**Note No:2**

Please check.

**Submitted by:**

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**On:**25/11/2024(18:42:32) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950)

**Note No:3**

Proposed LMI is found in line with OD.

Necessary approval may please be accorded by competent authority.

**Submitted by:**

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**On:**26/11/2024(14:30:04) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950)

**Note No:4**

Checked and forwarded.

**Submitted by:**

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**On:**01/12/2024(14:27:46) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:5**

Please review

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**On:**02/12/2024(17:28:19) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:6**

As discussed please review.

**Submitted by:**

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**On:**23/12/2024(21:25:04) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:7**

Reviewed and updated ,updated document attached as LMI Barring of Turbine Generators and Prevention of stalling updated

**Submitted by:**

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**On:**26/12/2024(04:43:01) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:8**

Attached LMI "LMI Barring of Turbine Generators and Prevention of Stalling Updated.pdf" reviewed.

Forwarded for approval.

**Submitted by:**

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**On:**26/12/2024(06:06:33) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:9**

Forwarded for approval

**Submitted by:**

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**On:**26/12/2024(09:29:52) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:10**

LMI may be reviewed pl

**Submitted by:**

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**On:**26/12/2024(18:34:53) **Ref:**SMTTPP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:11**

**Submitted by:**

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**On:**26/12/2024(19:26:15) **Ref:**SMTTPP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:12**

Proposed LMI found in line with the respective OD. Approval may kindly be accorded by competent authority.

**Submitted by:**

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**On:**28/12/2024(18:30:19) **Ref:**SMTTPP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:13**

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**On:**30/12/2024(16:33:34) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:14**

Checked and submitted for approval please

**Submitted by:**

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**On:**31/12/2024(09:51:28) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:15**

Please review.

**Submitted by:**

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**On:**31/12/2024(19:18:07) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:16**

It is requested to modify as per standard format of Simhadri LMI (Guidelines for Issue of LMIs as a Controlled Document) LMI/OGN/GEN/002 Rev:04 as uploaded on Simhadri EEMG LAN

**Submitted by:**

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**On:**01/01/2025(17:58:33) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:17**

LMI Updated as per Standard format of Simhadri LMI(Guidelines for issue of LMIs as a controlled Document) and Uploaded as **LMI Barring of Turbine Generators and prevention of Stalling(New Format).**

Please review and forward for Approval

**Submitted by:**

किल्लि तिरुमल राजशेखर - उप महाप्रबंधक

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**On:**11/01/2025(14:22:59) **Ref:**SMTPP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:18**

LMI reviewed

**Submitted by:**

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**On:**13/01/2025(12:08:05) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:19**

LMI is in line with the OD (OD-OPS-SYST-001\_Rev5\_BARRING OF TURBINE GENERATORS AND PREVENTION OF STALLING).

LMI may please be approved.

**Submitted by:**

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**On:**20/01/2025(11:02:46) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:20**

**Submitted by:**

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**On:**21/01/2025(15:56:56) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**Note No:21**

**Approved & Submitted by:**

समीर शर्मा - मुख्य महाप्रबंधक

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**On:**21/01/2025(16:20:18) **Ref:**SMTTP/OPN (MAIN PL/2024-25/OPERATIO/699950

**NTPC**

**SIMHADRI**

**LOCAL MANAGEMENT INSTRUCTION**

**Barring of Turbine Generators and Prevention of Stalling**

**LMI/OD/OPS/SYST/001 Rev.: 08**

**NOV -2024**

**Approved for Implementation by HOP (SIMHADRI)**

## LMI APPROVAL DOCUMENT

|    |                     |                                                                                                            |
|----|---------------------|------------------------------------------------------------------------------------------------------------|
| 1  | LMI NAME            | Barring of Turbine Generators and Prevention of Stalling                                                   |
| 2  | LMI No.             | LMI/OD/OPS/SYST/001 Rev:08                                                                                 |
| 3  | Reference Document  | COS-ISO-00-OD/SYST/001, Rev. No. : 5 Sep 2024. (Barring of Turbine Generators and Prevention of Stalling). |
| 4  | Superseded Document | LMI/OD/OPS/SYST/001 Rev:07 Dec 2022                                                                        |
| 5  | Prepared By         | Name: Killi Tirumala Rajasekhar<br>DGM(OPN)                                                                |
| 6  | Checked By          | Name: Somnath Bhattacharya<br>AGM (OPN)                                                                    |
| 7  | Checked By          | Name: Nilanjan Mandal<br>AGM (TMD)                                                                         |
| 8  | Checked By          | Name: Jayamol Rachel Varughese<br>AGM (C&I)                                                                |
| 9  | Checked By          | Name: Om Prakash<br>AGM (Maintenance)                                                                      |
| 10 | Checked By          | Name: Navin Kishore<br>AGM (EEMG)                                                                          |
| 11 | Reviewed By         | Name: Prasanta Kumar Jena<br>G M (O&M)                                                                     |
| 12 | Approved By         | Name: Sameer Sharma<br>H O P (Simhadri)                                                                    |

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|               |                                                                 |
|---------------|-----------------------------------------------------------------|
| <b>TITLE</b>  | <b>Barring of Turbine Generators and Prevention of Stalling</b> |
| <b>LMI No</b> | <b>LMI/OD/OPS/SYST/001 Rev:08</b>                               |

## 1.0 INTRODUCTION:

In Steam turbines, Barring gear arrangement is provided to keep it rotating at a nominal speed for preventing thermal bending (Eccentricity) in hot conditions. In case of turbines provided with **Hydraulic barring**, it has been seen that, post unit trip, when turbine fails to freely rotate on barring gear (barring gear stalling) it takes about 48 to 72 hours for Turbine rotor to become free, thus causing availability loss. The turning gear speed is about 80 rpm without vacuum and up to 130 rpm with vacuum. This speed varies from unit to unit.

This LMI draws attention towards operational precautions which are required to be taken to Prevent Barring stalling under all operating conditions and avoiding permanent bending, in steam turbines. It also guides in analyzing the reasons and taking required corrective actions.

## 2.0 SUPERSEDED DOCUMENTS & REVISION NOTES:

Superseded document: LMI/OD/OPS/SYST/001-Rev No-5, Issue 1 Dt. June 2020  
Rev03 Sept'2014: In line with Revised OD, LMI revised. Rev04 Jan'2020: Revision notes introduced in paragraph 2.0.

Rev05 June'2020: Revised in line with Revised OD May 2020.

Actions to be taken during TG capital OH & recently issued CC-OS IOM guidelines for prevention of stalling have been also included in this document. (Paragraphs 4.2.10, 4.2.1.1.1, 4.2.1.1.3, 4.2.1.1.4, 4.2.1.3.2, 4.2.2.2, 4.2.2.3, 4.2.2.4, 4.2.2.5, & 4.2.6.1)

Rev:06 May2022: revision in line with Revised OD April-2022

4.1 e added, ii)4.2.1.3.1,4.2.2.1,4.2.2.11 added; iii)4.2.4.2(e) added

iv)4.2.5.2,4.2.5.7,4.2.9 are modified also

Rev:06 May2022: revision in line with TMD's comments.

4.2.2.6, ii ) 4.2.31 iii) 4.2.4.d iv) 4.2.4.2.g, v) 4.2.5.4,vi) 4.2.6.3,vii) 4.2.7.5, viii) 4.2.10 are modified Rev: 07 Dec 2022: LMI review period changed from 3 years to 2 years

### **Rev: 08 Nov 2024:Revised in line with Revised OD Sep 2024**

i)4.1 f added

ii)4.2.1.1.5 modified(seal steam trap checking frequency added)

iii)4.2.1.3.2 Recording of JOP back pressure changed to every 30 days from 15 days

iv)4.2.4.1 e Ensure MS/CRH/HRH line is in depressurized condition added

v)4.2.2.1 Incorporation of alarm of 1st st pr > 2ksc with turbine trip included

vi)4.2.6.3 added

vii)Proper instruction of stoppage of oil pumps in case of requirement is mentioned

viii)Annexure-III check list added to counter reasons of barring and take corrective action in case barring stalling takes place

ix)4.2.2.10 Monitoring of brg temp during coasting down of turbine included and monitoring of LO pr/temp,LO pr after barring gear valve,Condensor vacuum included

x)4.2.4.2 j added

xi)4.2.1.1.4 Seal steam supply control valve to close on interlock if seal steam temperature < 280 deg(changed from temp < 260 deg)

|               |                                                                 |
|---------------|-----------------------------------------------------------------|
| <b>TITLE</b>  | <b>Barring of Turbine Generators and Prevention of Stalling</b> |
| <b>LMI No</b> | <b>LMI/OD/OPS/SYST/001 Rev:08</b>                               |

### 3.0 SCOPE:

This LMI is applicable to all four 500MW units of Simhadri STPS (Stage-I&II), which are of KWU Design steam turbines provided with Hydraulic Barring Gear manufactured by BHEL having see through type glands.

### 4.0 INSTRUCTIONS

#### 4.1 Factors Effecting Barring Stalling: -

Smooth barring gear operation is dependent on availability of the following

- a) Adequate lift by jacking oil at bearings.
- b) Adequate Radial and Axial clearances between Turbine shaft and Stationery components (e.g. Inter stage / Gland seals, Oil Guards / Catchers / Shields / Baffles).
- c) Quality of Seal steam, at rated temperature and pressure without any moisture / condensate.
- d) Lubricating oil at Operating temperature for driving the hydraulic wheel.
- e) Ensure no change in load on TG deck like CW trip etc.
- f) Critical operation parameters hotwell level, 1st stage pressure after unit trip, HP-IP Casing top- bottom temperature difference, LPT exhaust hood temperature should be within limit

Discrepancy in any of the above factors has been seen to affect the smooth barring operation. Barring operation is affected not only by the prevailing conditions but also by the operating parameters prior to tripping of turbine

#### 4.2 Actions needed to be taken under various operating conditions:

- 4.2.1 Machine is On-Load
- 4.2.2 coasting down when Turbine is coming to Barring speed.
- 4.2.3 During Grid failure/Total AC failure
- 4.2.4 During shut down, Turbine is on Barring gear
- 4.2.5 Turbine speed slowdown is observed during barring gear operation
- 4.2.6 Turbine comes to standstill
- 4.2.7 During startup
- 4.2.8 Vibration at Barring speed
- 4.2.9 Care during Post Barring stalling
- 4.2.10 During Unit Capital Overhauling
- 4.2.11 Recommissioning Checks after Overhauling
- 4.2.12 Others

|               |                                                                 |
|---------------|-----------------------------------------------------------------|
| <b>TITLE</b>  | <b>Barring of Turbine Generators and Prevention of Stalling</b> |
| <b>LMI No</b> | <b>LMI/OD/OPS/SYST/001 Rev:08</b>                               |

#### **4.2.1 WHEN MACHINE IS ON LOAD:**

##### **4.2.1.1 Ensure a condensate free auxiliary seal steam**

**4.2.1.1.1** Auxiliary seal steam temperature (prior seal steam supply station) should always be maintained above 290°C and Motorized drain valve MAW-10-AA0011 is intermittently opening /closing at **280/300Deg C** respectively

Resp:UCE/Turbine Desk Engineer

**4.2.1.1.2** If Direct source of auxiliary steam from CRH is closed condition and auxiliary seal steam for each unit is charged through Unit HT Aux. Steam header where it is charged from Station HT header and parameters are maintained at rated pressure i.e. 16 kg/cm<sup>2</sup> (g) and rated temperature 320°C. However, in case of new units (i.e. 500 MW/Group-2/3/4/5 as mentioned in Annexure 1), during initial startup, when aux steam is not available from other unit, then AUX PRDS may be charged from Main steam & maintain required seal steam parameters. But at higher pressure and temperature, it may be difficult to maintain the required steam parameters. In such case, after availability of required steam parameters through MS PRDS / other units, only CRH Source is to be lined up for HT header and auxiliary seal steam supply and all spray valves should be kept closed.

Resp:UCE/Turbine Desk Engineer

**4.2.1.1.3** Wherever one 500 MW unit has been added to existing units, for providing auxiliary seal steam to that particular unit at the time of trip, additional 50 NB (with Motorized valve) drain in the header needs to be provided at PRDS floor, which shall open on tripping of the additional unit, for supply of auxiliary seal steam

Resp:AGM (TMD)

**4.2.1.1.4** Seal steam supply control valve to close on interlock (for prevention of water entry if seal steam temperature <250 Deg C. If HP shaft temperature is more than 300 Deg C, Seal steam supply control valve to close on interlock if seal steam temperature <280 Deg C

Resp: AGM (C&I)

**4.2.1.1.5** All Steam traps on drains of Aux. PRDS / HT Header / Station HT header / Aux. seal steam line until auxiliary seal steam supply isolating valve (upstream of seal steam station) are functional and serviced annually. All traps to be checked jointly by TMD & operation and joint protocol may be made on a quarterly basis.

Resp: AGM (TMD)/ AGM (O)

|               |                                                                 |
|---------------|-----------------------------------------------------------------|
| <b>TITLE</b>  | <b>Barring of Turbine Generators and Prevention of Stalling</b> |
| <b>LMI No</b> | <b>LMI/OD/OPS/SYST/001 Rev:08</b>                               |

**4.2.1.1.6** Continuous draining through warm up orifice/Manual valve provided in Auxiliary Seal steam supply line, upstream of seal steam control valve is effective and helping to maintain required seal steam temperature. Otherwise, orifice size is required to be increased by at least 25%, in consultation with OS-ST.(Orifices are replaced with manual valve for better regulation)

Resp: AGM (TMD)/ AGM (O)

**4.2.1.1.7** In Turbine seal steam header, continuous warm up flow is taking place through given orifice (Parallel to MAL-81). Using non-contact thermometer ensures that temperature of piping near orifice is at least 100°C+. Otherwise, reasons for less temperature / flow through this line needs to be investigated e.g. Line/orifice choked etc. and corrected

Resp: AGM (TMD)/ AGM (O)

**4.2.1.2 Ensure that Turning gear valve supply is "on" and SLC turning gear is 'ON'**

Resp:UCE/Turbine Desk Engineer

#### **4.2.1.3 For healthy Jacking oil System**

**4.2.1.3.1** Ensure that the Jacking oil pump discharge filter has been cleaned before restart of the Unit, wherever provided, otherwise it may reduce the header pressure. Ensure DP across jacking oil filter should not be more than 5 to 10 ksc. Even after filter cleaning If DP is maintained high, then change the filter element

Resp: AGM (TMD)/ AGM (O)

**4.2.1.3.2** Ensure that the jacking oil back pressure of each bearing is recorded and monitored every **30 days**. As a preventive measure, this will help in detecting any possibility of Hose leakage/Puncture/Joint looseness/pressure gauge defects.

Resp: AGM (TMD)/ AGM (O)

**4.2.1.3.3** Ensure that the handles of Jacking oil supply valves of Generator Bearings 5,6,7 are in removed condition (Being in pathway, valves are liable to accidental closure / disturbance).

Resp: AGM (TMD)/ AGM (O)

|               |                                                                 |
|---------------|-----------------------------------------------------------------|
| <b>TITLE</b>  | <b>Barring of Turbine Generators and Prevention of Stalling</b> |
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#### 4.2.2 DURING COASTING DOWN

**(After Unit trip when turbine is approaching barring speed ensure that)**

**4.2.2.1** Opening of HPT casing drain & HPT evacuation valve as per logic to be ensured for depressurization of HPT casing. Monitor HPT 1st stage pressure and closing of CRH NRV. Ensure HPT 1st stage pressure should be less than 1.0 ksc. An alarm on first stage pressure > 2 Ksc with turbine in trip condition to be provided (As per CC OS IOM dt 14.02.2022)

**4.2.2.2** AC Jacking oil pump has taken start, {when speed is less than 510 rpm} and Jacking oil header pressure is above 140 kg/cm<sup>2</sup>. If it is less than 120 kg/cm<sup>2</sup>, then DCJOP can be taken in service. However, in such condition, header pressure should not be allowed to exceed 170 kg/cm<sup>2</sup>. If Jacking oil pressure is maintained less (with one pump), then reason to be analyzed and system to be checked for any leakages, passing of safety valves, pressure regulating valves, startup vents etc.

Resp:UCE/Turbine Desk Engineer

**4.2.2.3** Gate Valve gearing (Turning oil valve) is opened at speed less than 210 rpm. Take care that valve will close on protection, If Lube oil pressure after filter is less than 1.2 kg/cm<sup>2</sup>. **Oil Pressure after the barring gear valve to be monitored (both in Local & UCB)**

Resp:UCE/Turbine Desk Engineer

**4.2.2.4** Raise the lube oil temperature set point to 58 Deg C and maintain lube oil temperature from 55~ 60 Deg C.

Resp:UCE/Turbine Desk Engineer

**4.2.2.5** Maintain hot well at normal level with minimum variation. (Max +/-20 mm)

Resp:UCE/Turbine Desk Engineer

**4.2.2.6** Both CW Pumps to be kept in service and in no case CW pumps to be stopped or water box to be drained. No cooler / equipment should be drained which will affect the load on TG deck

Resp:UCE/Turbine Desk Engineer

**4.2.2.7** Trend of auxiliary seal steam, leak / seal steam header temperature does not indicate any sudden dip or decrease in temperature. Auxiliary seal steam temperature is maintained more than 290°C.

Resp:UCE/Turbine Desk Engineer

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**4.2.2.8** Due to the changeover of seal steam source from self-sealing to Auxiliary supply, seal steam header pressure remains positive (otherwise atmospheric air will rush in and cause chilling of glands causing barring seizure) and controller is maintaining the same on auto

Resp:UCE/Turbine Desk Engineer

**4.2.2.9** Record coasting down time from Unit trip to barring speed of 210 rpm & 120 rpm and compare with previous value and if changes are beyond 15%, investigate reasons.

Resp: AGM (TMD) / AGM (O)

**4.2.2.10** Monitor all bearing temperatures during coasting down on turbine, root cause analysis for any abnormal rise in bearing temperature to be carried out and to be rectified.

Resp: AGM (TMD) / AGM (O)

**4.2.2.11**

- Monitor Turbine speed along with related parameters (Annexure-II), mainly
- Hot well level-normal
- 1st stage pressure & CRH NRV close feedback
- Deaerator Level-Normal
- HP & IPT casing Top / Bottom temperature
- HPT exhaust temperature
- IP casing top / bottom temperature-Front and Rear
- LPT exhaust hood temperature
- Auxiliary Seal steam temperature (at seal steam control station-El. 8.5 m)
- Jacking oil Header pressure
- Generator Seal Oil pressure
- Seal steam pressure
- Seal steam temperature
- Lub oil pressure & temp
- LO pressure after barring gear valve
- TG bearing temperature
- Condenser vacuum

**4.2.2.12** In case of ACJOP is not available due to any reason DCJOP can be run continuously

Resp:UCE/Turbine Desk Engineer

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#### **4.2.3 ADDITIONAL ACTIONS DURING GRID/TOTAL AC FAILURE: ENSURE THAT**

**4.2.3.1** DC EOP & DC JOP to be started, in case the emergency Bus AC AOP and AC JOP fail to come in service

Resp:UCE/Turbine Desk Engineer

**4.2.3.2** Auxiliary seal steam supply station is urgently isolated by closing the manual isolating valve.

Resp:UCE/Turbine Desk Engineer

**4.2.3.3** Manual isolating valves of HP Bypass, Superheat spray, Reheat spray station is closed immediately

Resp:UCE/Turbine Desk Engineer

**4.2.3.4** Gland steam condenser exhaustor is re-started only when seal steam charging is to be done again

Resp:UCE/Turbine Desk Engineer

**4.2.3.5** Once the power supply is restored, Lubricating oil Temperature controller has been put in Auto. This will avoid excessive dropping of Lubricating oil temperature

Resp:UCE/Turbine Desk Engineer

**4.2.3.6** On power restoration, Condenser CW valves be immediately closed to prevent Condenser water box draining

Resp:UCE/Turbine Desk Engineer

**4.2.3.7** Ensure that after the restoration of power supply, open command is issued to vacuum breaker before starting control fluid pump

Resp:UCE/Turbine Desk Engineer

**4.2.3.8** Ensure that Turbine area MS / HRH / CRH piping / strainer drains to atmosphere / Unit flash tank are open and drains to Steam drain flash tank are closed. Ensure that manual isolating valves of MAL valves are actually closed

Resp:UCE/Turbine Desk Engineer

**4.2.3.9** If no AC AOP is running and Turning gear Valve Motor supply board has not yet been charged through emergency supply, then to prevent overloading of DCEOP, turning gear valve is closed manually. Ensure that Turning gear valve is opened after supply restoration & running of AC AOP

Resp:UCE/Turbine Desk Engineer

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#### 4.2.4 DURING SHUT DOWN, TURBINE IS ON BARRING GEAR

**During Shut down, when turbine is stabilized on barring gear, after 60 minutes of trip, ensure that:**

- Manual isolating valves of HPBP spray valves are closed, CRH line drains (upstream and downstream of NRV) are open.
- CRH NRV is closed and CRH NRV drain pot, drain valves are functioning as per logic.
- Hot well level is normal. Ensure that emergency make up valve is isolated.
- Lubricating oil temperature set point (55-60 deg) is on Auto and SLC AOP is on.
- No abnormal sound is coming near Glands / bearings and no abnormal vibration trend is observed. Check for reasons.
- No hammering is taking place in Turbine area piping. Take a walk down survey within 60-90 minutes of unit trip, when turbine has stabilized on barring gear.
- Manual isolating valve of MAL valves are physically open.

Resp:UCE/Turbine Desk Engineer

**4.2.4.1 If Unit is to remain out for more than 4 to 6 hours : Vacuum should preferably be killed after stabilization of the barring speed for around 2 hrs .To prevent air chilling of glands, vacuum killing is to be carried out with following Sequence of operation**

- Stop Vacuum pump.
- Kill Vacuum by opening Vacuum breaker
- Isolate auxiliary Gland seal steam to Turbine at Vacuum of 0.2 kg/cm<sup>2</sup> (or 152 mm Hg) by closing manual isolating valve. Additionally Leak off steam Bypass (Motorized) valve is to be kept open. In case of any accidental moisture / condensate entry into auxiliary seal steam piping, this opening will continue to drain / purge any condensate and accidental entry of moist / condensate into turbine glands can be prevented. At the time of start up, after seal steam charging, this motorized valve needs to be closed.
- Stop gland seal steam Exhauster immediately after isolation of Auxiliary gland seal steam.
- Open the Turbine MS / HRH / CRH line / strainer drains to atmosphere / Unit flash tank and isolate drains to Steam drain flash tank. Ensure MS/CRH/HRH line is in de-pressurized condition.
- Isolate Aux. steam and seal steam supply to both the TDBFP turbines

Resp:UCE/Turbine Desk Engineer

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#### 4.2.4.2 If Unit is to be taken under long shut down:

- a) Check that Turbine area MS / HRH / CRH piping / strainer drains to atmosphere / Unit flash tank are open and drains to Steam drain flash tank are closed. Ensure that manual isolating valves of MAL valves are actually closed.
- b) Monitor the trend of parameters (Annexure-II) and check for any exception.
- c) Check that Deaerator pegging steam is isolated and check that no passing is taking place
- d) Check that steaming is not taking place while Deaerator draining. Before CW stoppage and Condenser water box draining, ensure that Deaerator level is brought to normal level so that its overflow does not spill over to hotwell and causes unwanted steaming.
- e) When vacuum is killed ensure that LPT exhaust hood spray and its bypass valves are in close condition and ensure no passing from these valves.
- f) Subsequent to CW stoppage, if any Deaerator draining is required and feed water is hot, it is required to be drained through drain connected to Unit flash tank. However there is possibility of the excess water / wet steam entering back into the MS / HRH strainer / piping, which is in hot condition and can affect MS / HRH strainer. {For preventing the same, drain (upstream of Unit flash tank) be routed to atmosphere at a safe location such as A-row drain, so that deaerator can be drained through atmosphere and not through Unit flash tank}.
- g) One or both CW water boxes to be drained / emptied, if required for taking up of any maintenance jobs in CW system, preferably after 48 hours of Unit shut down once the HP casing/rotor temperature comes below 325 deg C. Till then keep at least 1 cw pump running.

In view of TG deck being mounted on springs, it has been observed that when water box gets emptied out in hot condition, adverse changes in clearances in LPT turbine takes place, which affect the smooth barring gear operation. Decrease in barring speed to be monitored while draining of condenser water boxes/CW ducts/stoppage of CW pump

- h) If control fluid pump is stopped, manual isolating valves of LP BP system injection water supply valve is to be closed in stage #1 units. During subsequent start up, normalization of the same is to be ensured.
- i) Ensure that Boiler stop valves are closed and power supply is isolated.
- j) To meet the any exigency requirements, if draining of CW duct/condenser water box/stoppage of any or both CW pumps is required with HP casing/rotor temp > 325 deg, it may be carried out after due diligence & mutual concurrence between site & COS-ST on case to case basis

Resp:UCE/Turbine Desk Engineer

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#### **4.2.5 TURBINE SPEED SLOW DOWN IS OBSERVED DURING BARRING GEAR OPERATION:**

**4.2.5.1** Check that Turning gear valve has actually opened and ensure that hydraulic oil pressure is 6.0 kg/cm<sup>2</sup> + after barring gear valve.

Resp:UCE/Turbine Desk Engineer

**4.2.5.2** Check that adequate jacking oil pressure is available in header (130-140 kg/cm<sup>2</sup>). If it is low, say 130 kg/cm<sup>2</sup>, then second JOP is to be taken into service. Passing of JOP discharge pressure relief valves, startup vents, and JOP header PRV's if any to be checked. Then JOP Changeover, PRV adjustment etc. to be taken up to address the defects

Resp:UCE/Turbine Desk Engineer

**4.2.5.3** Check that Lubricating oil temperature is 45°C and raise to 55°C-60°C gradually and observe the behavior for any improvement. Due to lower viscosity and lower coefficient of friction, higher lubricating oil temperature shall help in improving the barring speed.

Resp:UCE/Turbine Desk Engineer

**4.2.5.4** Check that Hot well level is normal. If still barring speed is in decreasing trend, reduce hot-well level by 150mm below normal level (this check has been seen to help at number of stations).

Resp:UCE/Turbine Desk Engineer

**4.2.5.5** Take second AOP in service when barring speed drops to 100 rpm with vacuum. This shall help in increasing oil pressure to Hydraulic barring wheel. After that turning gear valve can be attempted for further opening manually, so that any margin improves the Turning wheel torque.

Resp:UCE/Turbine Desk Engineer

**4.2.5.6** Check that jacking oil pressure is actually available to all the bearings. Check, record and Compare jacking oil pressure with previously set values.

Resp:UCE/Turbine Desk Engineer

**4.2.5.7** If auxiliary seal steam is charged, check that seal steam temperature is at least 280°C and Seal steam header pressure is normal. If difficulty is faced in achieving required temperature, then in order to avoid condensation / chilling due to low temperature, sufficient flow of auxiliary seal steam can be ensured by opening both i.e. seal steam supply control valve/motorized bypass valve and leak off control valve/motorized bypass valve. This will help in maximizing seal steam flow while maintaining seal steam header pressure.

Resp:UCE/Turbine Desk Engineer

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**4.2.5.8** If condenser is under vacuum and seal steam temperature is not under control, then kill the condenser vacuum and isolate seal steam supply and stop the GSC vapour exhauster.

Resp:UCE/Turbine Desk Engineer

**4.2.5.9** Check the trend of parameters e.g. seal / leak steam header temperature, hot well level, LPT exhaust hood temperature, HP, IP casing top / bottom temperature, CW water box pressure etc. (Annexure-II) and take corrective action for abnormality. (Reg. Steam parameters, PI Refer Annexure I).

Resp:UCE/Turbine Desk Engineer

**4.2.5.10** Check shaft vibrations for any abnormal increase / change, it may help in indicating the zone, where free rotation of the rotor is getting resisted.

Resp:UCE/Turbine Desk Engineer

**4.2.5.11** Observe drop / record drop in MOT level due to opening of gate valve gearing.

Resp:UCE/Turbine Desk Engineer

**4.2.5.12** Check LPT exhaust hood spray valve is closed and no passing is there

Resp:UCE/Turbine Desk Engineer

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#### **4.2.6 TURBINE ROTOR COMES TO STANDSTILL:**

**4.2.6.1** Then hand barring should be done immediately. If rotor is freely rotating with one person, then effort should be taken to put to bring back the TG on barring gear. Rotate half a turn (180°) every 10 minutes (Make a mark on the shaft near Generator front). If rotor is rotating tight by one person, then don't put extra force to rotate the rotor.

Resp AGM (TMD)/SCE/UCE/Turbine Desk Engineer

#### **4.2.6.2** Check for abnormality in parameters as per Annexure-II

Resp:UCE/Turbine Desk Engineer

**4.2.6.3** Even if rotor freeness is not increased and machine cannot be put on barring, in that case condenser vacuum may be killed as per procedure

Resp:UCE/Turbine Desk Engineer

**4.2.6.4** If Barring gear is not able to rotate by one person, then no extra force on rotor should be applied and it should be allowed to cool, till the time it becomes free. HP turbine control valves should never be opened which can cause speed to rise by steam force. In this condition if any extra force is applied, it will damage the seals (bending, wearing, dislodging) and adversely affect the turbine clearances, efficiency and reliability. Keep on trying with one person, every 15 minutes. Check for preventive actions advised at Sl. No.4.2.5. If with passage of 8-12 hours, no improvement is seen towards freeness of rotor, then fast cooling can be started (in consultation with CC-OS-ST and TC Doc no. OGN-OPS- MECH-042).

Resp: AGM (O) / AGM (TMD)

**4.2.6.5** In case of failure of lube oil system or jacking oil system, if rotor has come to rest, no attempt should be made to turn the shaft either by hand or power until oil supplies are restored.

Resp: AGM (TMD)/ AGM (O)/UCE/Turbine Desk Engineer

**4.2.6.6** In case of failure of seal oil supplies and if it is essential to turn the rotors, only hand barring Should be used

Resp: AGM (TMD)/ AGM (O)/UCE/Turbine Desk Engineer

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#### 4.2.7 DURING START UP:

**4.2.7.1** If seal steam supply has been isolated, then before opening manual isolating valve, it should be ensured that Warm up valve is open to atmosphere and seal steam supply motorized Bypass valve is closed.

Resp:UCE/Turbine Desk Engineer

**4.2.7.2** If HP shaft temperature is above 300°C then before seal steam charging, ensure that Seal steam temperature is above 285°C, otherwise Seal steam valve shall not be able to open (as per logic).

Resp:UCE/Turbine Desk Engineer

**4.2.7.3** When HP shaft temperature is below 300°C, then ensure that seal steam temperature is about 230-240°C (greater than 210°C).

Resp:UCE/Turbine Desk Engineer

**4.2.7.4** Ensure that drains of LPBP line and HP / IP Stop / Control valve and drain valve **MAL-81** are open, otherwise seal steam header temperature may fall due to entry of moist steam into the seal steam leak off header

Resp:UCE/Turbine Desk Engineer

**4.2.7.5** At the time of HP bypass charging it should be ensured that CRH NRV remains fully close and monitor HPT 1st pressure. If not fully closed, it may cause speed slow down / Barring stalling/quenching of HPT casings. CRH NRV to be ensured closed from local. Ensure that oil supply of CRH NRV servomotor supply is normal.

Resp:UCE/Turbine Desk Engineer

**4.2.7.6** Check that Condenser CW is charged gradually to prevent disturbance and sudden shock to Condenser (in hot condition) due to starting of CW pumps. Initially Condenser water box vents should remain open (in case condenser water box is not provided with Air release valves) and Condenser inlet valves are to be opened only 10%, with outlet valves remaining full open. Subsequently after observing water from vents, CW inlet and outlet valves are to be opened fully and then pump can be taken in service.

Resp:UCE/Turbine Desk Engineer

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#### **4.2.8 VIBRATION AT BARRING SPEED:**

If there is a abnormal sound or increasing vibration shaft / bearing vibration trend, then with continued rub there is a probability of bend getting developed which can result in high vibrations during subsequent start up. In such case, barring gear should be stopped and hand barring should be done with half a revolution every 10 minutes. Once the trend reverses or cause has been rectified, the turbine can be put again on barring gear operation. However in such case turbine should be kept at barring gear for at least 8 hours. (Reg. Steam parameters,PI Refer Annexure 1).

Resp: AGM (TMD)/ AGM (O)/UCE/Turbine Desk Engineer

#### **4.2.9 CARE DURING POST BARRING STALLING:**

As a consequence of failure of barring gear and subsequently after restoration turbine needs to be rotated at barring gear for at least 6 to 8 hours and see that eccentricity / shaft vibration trend has flattened out. Otherwise, since rotor eccentricity is increased due to bowing, it will result in high vibrations especially during transient / start up when passing through critical Speed. If JOP shaft lift got disturbed, then JOP shaft lift to be readjusted only after putting TG on barring gear for at least 2 hours.

Resp: AGM (TMD)/ AGM (O)/UCE/Turbine Desk Engineer

#### **4.2.10 Activities/Jobs to be taken up during TG capital overhaul to avoid barring gear stalling.500MW KWU NEW VERSION MACHINE:**

- a) Modified procedure is to be adapted for centering of HPT inner casing w,r,t. outer casing as per latest CC-OS Circular

Resp:AGM(TMD)

- b) HPT gland radial clearances are to be set 0.5 to 0.6mm

Resp:AGM(TMD)

- c) In case of see-through seals at HPT front, 0.3mm feeler of length 400mm go to be ensured all around the seals and 0.3mm feeler of length 70mm(as per drawing) go to be ensured all around the seals at IPT rear.

Resp:AGM(TMD)

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#### **4.2.11 Recommissioning After Overhaul:**

**Before putting TG set on turning gear after overhaul following, must be ensured**

**4.2.11.1** The lube oil system is normalized and in circulation for more than 4 hours at-least without jacking oil filter and Lub oil filter getting choked, and there is no chokage of line filters.

Resp: AGM (O)/ AGM (TMD)

**4.2.11.2** The lube oil line throttles at the inlet of bearings are adjusted to the initial set values and a protocol is signed by Turbine Maintenance and recommissioning.

Resp: AGM (O)/ AGM (TMD)

**4.2.11.3** Shaft lifts are adjusted and protocol is signed.

Resp: AGM (O)/ AGM (TMD)

**4.2.11.4** Jacking oil header pressure is set at 140 to 170 kg/ cm<sup>2</sup>

Resp:AGM(O)/AGM(TMD)

**4.2.11.5** The relief valves in jacking oil lines are set as recommended and protocol is signed.

Resp AGM (O)/ AGM (TMD)

**4.2.11.6** After all adjustment lube oil system shall be established and return oil flow through all the bearings shall be checked without jacking oil pump.

Resp: AGM (O)/ AGM (TMD)

**4.2.11.7** All SGC oil system and SGC seal oil to be proven and protocol is signed by recommissioning and C&I maintenance.

Resp: AGM (O)/ AGM (C&I)

**4.2.11.8** Availability of DG set, DC EOP,DC SOP and DC JOP is ensured.

Resp:UCE/Turbine Desk Engineer

**4.2.11.9** After putting the TG set on turning gear it will be kept under observation for at least 8 hrs and the bearing temperature, vibration, drain oil temperature shall be recorded every hour. These readings shall be compared with the initial reference readings and any deviation shall be analysed and corrective action taken before clearing for vacuum pulling.

Resp:AGM(O)/AGM(TMD)/AGM(C&I)

**4.2.11.10** All the above checks shall be approved by HOD(O) before clearing the TG set for rolling.

Resp:AGM(O)

**4.2.11.11** The copy of recommissioning protocols should be made available in control room for the reference of operation engineers.

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#### **4.2.12 MISCELLANEOUS:**

As mentioned at 4.2.4.3, deaerator drain to Unit flash tank (for 500 MW BHEL supplied turbines) be diverted to atmosphere at a safe location e.g. A-row drain so that it is possible to drain deaerator, without draining through Unit flash tank.

Resp:AGM(O)

#### **4.2.13 Stopping of barring gear and oil pumps**

After cooling of HP shaft & casing below 100 deg. Barring and oil can be stopped. Sometimes, In case of exigency or minor works towards generator side, there is a need to stop barring gear & oil at higher temperatures. To stop the barring gear and Lub & jacking oil at higher shaft temperature, matter to be discussed with COS-ST on case to case to basis. With clearance from COS and suggested precautions oil can be stopped.

#### **5.0 REFERENCES DOCUMENT:**

- 1.COS-ISO-00-OD/SYST/001, Rev. No. : 5 Sep 2024. (Barring of Turbine Generators and Prevention of Stalling).
- 2.500MW Turbine Manual of M/S BHEL.

#### **6.0 RECORDS**

The parameters are to be recorded as given in format (annexure-2), provided in each unit. All the related activities carried out are to be recorded in UCE / Turbine desk Engineer log book. Records must be kept for above activities as follows: Operational parameters by operation department.

Any deviation operational parameters to be highlighted by operation department, exception of the TG barring gear stalling by TMD.

#### **7.0 IMPLEMENTATION**

HOD (O) is responsible for implementation of this LMI

#### **8.0 REVIEW**

The HOD (O) will be responsible for reviewing this LMI on a 2-yearly basis.

#### **9.0 DISTRIBUTION LIST**

Simhadri Local Area Network

Resp:AGM(O)/AGM(TMD)

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## Annexure-I

**Guidelines for Seal Steam Temperatures for preventing Barring Stalling at Simhadri STPS Stage-I&II units:**

- The specified seal steam temperature is to maintain 290°C.

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**Annexure-II**  
**PARAMETERS TO BE MONITORED HOURLY**

| <b>S.No</b> | <b>Parameter</b>                                                |  |  |  |
|-------------|-----------------------------------------------------------------|--|--|--|
| 1           | Load                                                            |  |  |  |
| 2           | Speed                                                           |  |  |  |
| 3           | Vacuum                                                          |  |  |  |
| 4           | MS throttle pressure &HPT 1st stage pressure                    |  |  |  |
| 5           | MS temperature                                                  |  |  |  |
| 6           | CRH pressure                                                    |  |  |  |
| 7           | CRH temperature                                                 |  |  |  |
| 8           | HRH temperature                                                 |  |  |  |
| 9           | Deaerator level                                                 |  |  |  |
| 10          | Hotwell level                                                   |  |  |  |
| 11          | LPT exhaust hood temperature front                              |  |  |  |
| 12          | LPT exhaust hood temperature Rear                               |  |  |  |
| 13          | Condenser hot well temperature                                  |  |  |  |
|             | <b>Lub oil / Jacking oil / Seal oil parameters</b>              |  |  |  |
| 14          | Jacking oil header pressure                                     |  |  |  |
| 15          | Jacking oil pressure at Brg-5                                   |  |  |  |
| 16          | Jacking oil pressure at Brg-6                                   |  |  |  |
| 17          | Jacking oil pressure at Brg-7                                   |  |  |  |
| 18          | Lubricating oil pressure (Before filter)                        |  |  |  |
| 19          | Lubricating oil pressure (After filter)                         |  |  |  |
| 20          | Lubricating oil supply temperature                              |  |  |  |
| 21          | H2 & Seal oil pressure (TE/GE)                                  |  |  |  |
|             | <b>CW parameters</b>                                            |  |  |  |
| 22          | Condenser Cooling water pressure Inlet-pass-1                   |  |  |  |
| 23          | Condenser Cooling water pressure Inlet-pass-2                   |  |  |  |
| 24          | Condenser cooling water outlet temp.-pass-1                     |  |  |  |
| 25          | Condenser cooling water outlet temp.-pass-2                     |  |  |  |
|             | <b>Seal steam system Parameters</b>                             |  |  |  |
| 26          | Seal steam header pressure                                      |  |  |  |
| 27          | Aux seal steam temperature upstream of Seal steam control valve |  |  |  |

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| <b>S.No</b> | <b>Parameter</b>                                               |  |  |  |
|-------------|----------------------------------------------------------------|--|--|--|
| 28          | Seal steam valve opening ( %)                                  |  |  |  |
| 29          | Leak steam valve opening ( % )                                 |  |  |  |
| 30          | Seal/leak steam header temperature (front)                     |  |  |  |
| 31          | Seal/leak steam header temperature (Rear)                      |  |  |  |
| 32          | Gland steam exhauster status (On/off)? (PFT-Position feedback) |  |  |  |
|             | <b>Bearing metal temperatures</b>                              |  |  |  |
| 33          | Bearing-1 metal temperatures                                   |  |  |  |
| 34          | Bearing-2 metal temperatures                                   |  |  |  |
| 35          | Bearing-3 metal temperatures                                   |  |  |  |
| 36          | Bearing-4 metal temperatures                                   |  |  |  |
| 37          | Bearing-5 metal temperatures                                   |  |  |  |
| 38          | Bearing-6 metal temperatures                                   |  |  |  |
| 39          | Bearing-7 metal temperatures                                   |  |  |  |
|             | <b>Turbine vibrations</b>                                      |  |  |  |
| 40          | HPT front shaft vibn (X,Y )                                    |  |  |  |
| 41          | HPT rear shaft vibration (X,Y )                                |  |  |  |
| 42          | IPT rear shaft vibration (X,Y )                                |  |  |  |
| 43          | LPT rear shaft vibration (X,Y )                                |  |  |  |
| 44          | Generator front shaft vibration (X,Y )                         |  |  |  |
| 45          | Generator rear shaft vibration (X,Y )                          |  |  |  |
| 46          | Exciter shaft vibration (X,Y )                                 |  |  |  |
| 47          | HPT front Brg. vibn (X,Y )                                     |  |  |  |
| 48          | HPT rear Brg. vibration (X,Y )                                 |  |  |  |
| 49          | IPT rear brg. vibration (X,Y )                                 |  |  |  |
| 50          | LPT rear brg. vibration (X,Y )                                 |  |  |  |
| 51          | Generator front brg. vibration (X,Y )                          |  |  |  |
| 52          | Generator rear brg. vibration (X,Y )                           |  |  |  |
| 53          | Exciter brg. vibration (X,Y )                                  |  |  |  |
|             | <b>Expansions</b>                                              |  |  |  |
| 54          | Axial shift                                                    |  |  |  |
| 55          | HPT rotor expansion                                            |  |  |  |

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| <b>S.No</b> | <b>Parameter</b>                              |  |  |  |
|-------------|-----------------------------------------------|--|--|--|
| 56          | HPT casing Overall or DE.                     |  |  |  |
| 57          | LPT rotor expansion                           |  |  |  |
| 58          | IPT casing Overall or DE.                     |  |  |  |
|             | <b>Casing &amp; Differential Temperatures</b> |  |  |  |
| 59          | HPT Top / Bottom DT                           |  |  |  |
| 60          | IPT front Top / bottom DT                     |  |  |  |
| 61          | IPT Rear Top / Bottom DT                      |  |  |  |
| 62          | HP casing temp. (100%)                        |  |  |  |
| 63          | HP casing temp. (50 %)                        |  |  |  |
|             | <b>Flash tank temperatures</b>                |  |  |  |
| 64          | HP flash tank temperature                     |  |  |  |
| 65          | LP flash tank temperature                     |  |  |  |
| 66          | Steam drain flash tank temperature            |  |  |  |
| 67          | Unit Flash tank temperature                   |  |  |  |
|             | <b>Spray flows</b>                            |  |  |  |
| 68          | Superheater Spray flow                        |  |  |  |
| 69          | Reheater spray Flow                           |  |  |  |
| 70          | LPT Exhaust hood temp front                   |  |  |  |
| 71          | LPT Exhaust hood temp Rear                    |  |  |  |
| 72          | Condensor hot well temperature                |  |  |  |

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**Annexure-III****Checklist to prevent barring stalling & Corrective measures in case of stalling**

**Following is common “Checklist for Pre-requisites for TG Barring” that is to be fulfilled in all situations:**

| <b>S.No</b> | <b>Points to be observed</b>                                                                                                                                                                                                                                                             | <b>Checked (Yes/No)</b> |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 1           | Check Auxillary Seal Steam Temperature >290 deg C(or as per scheme mentioned by OEM)                                                                                                                                                                                                     |                         |
| 2           | Check that warm-up line upstream of Seal Steam Control Valve is Open and Orifice across MAL-81 functioning properly.                                                                                                                                                                     |                         |
| 3           | Following periodic checks to be done when unit is in Operation:<br>Temperature mapping of Seal Steam Warm-up line (every 7 days)<br>Back pressure of Turbine JOP line at each bearing (every 15 days)<br>JOP Filter dP which should be maintained below 10 ksc (Whenever JOP in service) |                         |
| 4           | Ensure that SLC for Lube Oil Pump is ON, Pump comes into service as per logic and develop desired pressure                                                                                                                                                                               |                         |
| 5           | Ensure that SLC for JOP and Barring Gear Valve is ON and Supply to respective drives is normal.                                                                                                                                                                                          |                         |
| 6           | Ensure that Hot-well Level is normal and is not disturbed (variation within +/- 20 mm)                                                                                                                                                                                                   |                         |
| 7           | Ensure that no CW Pump is switched off (both Pumps to be kept in service) and no CW flow is disturbed (No pump changeover is recommended)                                                                                                                                                |                         |
| 8           | Check Lube Oil temperature is maintained between 55-60 deg C                                                                                                                                                                                                                             |                         |
| 9           | Ensure that JOP kicks in at 510 rpm and develops header pressure > 140 ksc. In case AC JOP does not come into service or develop adequate pressure, DC JOP or 2 no's JOP is taken into service.                                                                                          |                         |
| 10          | Ensure that Turning Gear Valve opens at 210 rpm and develops hydraulic oil pressure after barring valve > 6 ksc ( check/record pressure drop). Check for full opening of Turning Gear Valve                                                                                              |                         |
| 11          | Ensure that after turbine trip, all Turbine Stop Valves and CRH NRV get fully closed and verify that 1 <sup>st</sup> Stage pressure < 1.0 ksc                                                                                                                                            |                         |
| 12          | Ensure that all drains operate as per logic                                                                                                                                                                                                                                              |                         |
| 13          | Check HP Turbine and IP Turbine Casing Metal Temperature Top / Bottom temperature for excessive difference                                                                                                                                                                               |                         |
| 14          | Check that Manual isolating valve for HPBP Spray, SH/RH Spray, LPBP Spray are closed                                                                                                                                                                                                     |                         |
| 15          | Check MS/HRH/CRH/Strainer drains are open to Atmospheric Flash tank and not connected to Condenser                                                                                                                                                                                       |                         |

**Annexure-III**

|               |                                                                 |
|---------------|-----------------------------------------------------------------|
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### Checklist for different Situations

#### Situation A: Turbine coasting down after Turbine Trip

| S. No | Points to be observed                                                                                                                                                                       | Checked (Yes/No) | Remarks |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------|
| 1     | Refer “ <b>Checklist for Pre-requisites for fulfilling criteria for TG Barring</b> ”                                                                                                        |                  |         |
| 2     | Check opening of HPT Evacuation line and HPT Casing drain                                                                                                                                   |                  |         |
| 3     | Monitor and Record Turbine coasting down period till 210 rpm & 120 rpm is achieved and compare with last observed values. This Coasting down period shall be within 15% of previous values. |                  |         |

#### Situation B: Grid failure or Total AC failure

| S.No. | Points to be observed                                                                                                        | Checked (Yes/No) | Remarks |
|-------|------------------------------------------------------------------------------------------------------------------------------|------------------|---------|
| 1     | Check Start of all DC drives – DC EOP, DC JOP, DC Seal Oil Pump, etc                                                         |                  |         |
| 2     | Refer “ <b>Checklist for Pre-requisites for fulfilling criteria for TG Barring</b> ” for applicable points                   |                  |         |
| 3     | Condenser CW Valves (if required manually) are to be closed to prevent draining of water from Condenser tubes and Water box. |                  |         |
| 4     | Ensure that Vacuum Breaker is opened in case power is restored                                                               |                  |         |

#### Situation C: Reduction in Turbine speed while coasting down

| S. No. | Points to be observed                                                                                      | Checked (Yes/No) | Remarks |
|--------|------------------------------------------------------------------------------------------------------------|------------------|---------|
| 1.     | Refer “ <b>Checklist for Pre-requisites for fulfilling criteria for TG Barring</b> ” for applicable points |                  |         |
| 2.     | Check for any abnormal sound at Turbine glands or excessive shaft vibrations                               |                  |         |
| 3.     | In case Turbine will be kept on barring for more than 6 hours, Open Vacuum breaker                         |                  |         |
| 4.     | Check isolation of Gland Seal Steam to Turbine                                                             |                  |         |
| 5.     | Check that GSC Exhauster fans are stopped                                                                  |                  |         |
| 6.     | Check isolation of Auxiliary Supply Steam to TDBFP and Seal Steam to Drive Turbine                         |                  |         |

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**Situation D: Reduction in Turbine speed while on Barring**

| <b>S. No.</b> | <b>Points to be observed</b>                                                                                                                                                                                                     | <b>Checked (Yes/No)</b> | <b>Remarks</b> |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|
| 1.            | Refer “ <b>Checklist for Pre-requisites for fulfilling criteria for TG Barring</b> ” for applicable points                                                                                                                       |                         |                |
| 2.            | If drop in JOP Header pressure is observed, check for passing of JOP Relief valve and/or Pressure Regulating Valve. Carry out any rectification as required                                                                      |                         |                |
| 3.            | In case Turbine barring keeps falling, reduce Hotwell level by 150 mm below normal level. This may help in improving Barring speed                                                                                               |                         |                |
| 4.            | Take 2 <sup>nd</sup> AOP in service to increase oil flow to Turning Wheel                                                                                                                                                        |                         |                |
| 5.            | Check and record JOP pressure for individual Turbine bearings and compare it with previous recorded values                                                                                                                       |                         |                |
| 6.            | Check the trend of parameters e.g. seal/leak steam header temperature, hot well level, LPT exhaust hood temperature, HP, IP casing top/bottom temperature, CW water box pressure etc. and take corrective action for abnormality |                         |                |
| 7.            | Check for any abnormal sound at Turbine glands or excessive shaft vibrations                                                                                                                                                     |                         |                |
| 8.            | In case Seal Steam Temperature is not under control, kill Condenser Vacuum                                                                                                                                                       |                         |                |
| 9.            | Check isolation of Gland Seal Steam to Turbine                                                                                                                                                                                   |                         |                |
| 10.           | Check that GSC Exhauster fans are stopped                                                                                                                                                                                        |                         |                |
| 11.           | Check isolation of Auxiliary Supply Steam to TDBFP and Seal Steam to Drive Turbine                                                                                                                                               |                         |                |

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**Situation E: When Turbine Barring Stalls**

| <b>S. No.</b> | <b>Points to be observed</b>                                                                                                                                                                                                                    | <b>Checked (Yes/No)</b> | <b>Remarks</b> |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|
| 1.            | Hand barring to be carried out immediately and rotate shaft by 180° every 10 min. In case rotor is free, try to put machine in Hydraulic barring                                                                                                |                         |                |
| 2.            | Check for any anomaly in important parameters that are likely to affect Barring                                                                                                                                                                 |                         |                |
| 3.            | Check that no excessive force is used to rotate Turbine manually and only one person shall carry out hand barring. In case, shaft does not rotate, let the shaft lay idle for few minutes and next attempt to be carried out in next 15 minutes |                         |                |
| 4.            | Check that rotor is not free even after 8-10 hrs elapse, proceed to carry out Controlled Fast Cooling as per TC Doc no. OGN-OPS- MECH-042                                                                                                       |                         |                |
| 5.            | Once machine is put in barring keep it such for 6-8 hrs to remove any Rotor sag                                                                                                                                                                 |                         |                |
| 6.            | If JOP shaft lift got disturbed, then JOP shaft lift to be re-adjusted only after putting TG on barring gear for at least 2 hours                                                                                                               |                         |                |

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**Situation F: During Start-up**

| <b>S. No.</b> | <b>Points to be observed</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Checked (Yes/No)</b> | <b>Remarks</b> |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|
| 1             | Check that admission of Seal Steam is done after proper monitoring of both Shaft Temperature and Seal Steam Temperature. {If HP shaft temperature is above 300°C then before seal steam charging, ensure that Seal steam temperature is above 285°C, otherwise Seal steam valve shall not be able to open (as per logic). When HP shaft temperature is below 300°C, then ensure that seal steam temperature is about 230-240°C (greater than 210°C)}                                                           |                         |                |
| 2             | Check open position of LPBP line drains and HP/IP Stop/Control valve drain and MAL- 81                                                                                                                                                                                                                                                                                                                                                                                                                         |                         |                |
| 3             | Check that CRH NRV is fully closed (both in DCS and also from Local) while charging of HP bypass and monitor HPT 1st Stage pressure.                                                                                                                                                                                                                                                                                                                                                                           |                         |                |
| 4             | Check that Condenser CW is charged gradually to prevent disturbance and sudden shock to Condenser (in hot condition) due to starting of CW pumps. Initially Condenser water box vents should remain open (in case condenser water box is not provided with Air release valves) and Condenser inlet valves are to be opened only 10%, with outlet valves remaining full open. Subsequently after observing water from vents, CW inlet valve are to be opened fully and then second pump can be taken in service |                         |                |

|               |                                                                 |
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**Situation G: During Shut-down for Overhauling**

| <b>S. No.</b> | <b>Points to be observed</b>                                                                                                                                                                          | <b>Checked (Yes/No)</b> | <b>Remarks</b> |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|
| 1.            | Refer “ <b>Checklist for Pre-requisites for fulfilling criteria for TG Barring</b> ” for applicable points                                                                                            |                         |                |
| 2.            | Check that Deaerator pegging steam is isolated and check that no passing is taking place                                                                                                              |                         |                |
| 3.            | Check that Deaerator level is brought to normal level before stopping CW pumps and draining of Condenser water box so that its overflow does not spills over to hotwell and causes unwanted steaming. |                         |                |
| 4.            | Deaerator draining of hot feedwater is to be done taking adequate precaution that steaming does not ingress by return flow into Strainer through Steam piping                                         |                         |                |
| 5.            | CW water boxes are to be drained/emptied, preferably after 48 hours of Unit shut down once the HP casing / rotor temperature comes below 325 Deg C.                                                   |                         |                |
| 6.            | If control fluid pump is stopped, manual isolating valves of LP BP system injection water supply valve are to be closed. During subsequent start up, normalization of the same is to be ensured       |                         |                |
| 7.            | Check that Boiler stop valves are closed and power supply is isolated                                                                                                                                 |                         |                |